

This is why vector graphics do not lose their quality and clarity on being scaled up and down because they are, in essence, mathematical equations. Instead of multiplying the equation by 2 like in our example, we can multiply it by any real number we want. Math you beautiful thing you!

Now that you have a gist of what they are, each vector can be customised to have a particular stroke colour, thickness, fill and shape while still maintaining a small file size.

The very first vector drawing program called Sketchpad was created in 1960 by Ivan Sutherland. It was revolutionary during the day and is even hailed as the ancestor to Computer Aided Design (CAD).

In the days of yore, all 2D computer graphics were vector graphics. Since then vectors have a reappearance in home gaming systems, arcade games and flash animations.

Here are some simple uses of a vector graphics editor:

Logos

Illustrator is always the first choice even though photoshop also has its place in logo design. Logos are meant to be reshaped and redesigned – vectors are the way to go. Ideal for a beginner.



A translation moves every point of a figure or a space by the same amount in a given direction

T-Shirts

Illustrator makes this an easier task with its clear vector images.

You get on your t-shirt, something pretty close to what you designed on your screen.

Business cards

The multiple artboards feature on Illustrator now means you don't need multiple files for each name.

What's the difference in software?

While comparing graphics editors, we have to take into consideration the range of file formats supported. If the image is stored in a bitmap format (the normal pixel format you use in photos, for example), it is called a bitmap editor (e.g. Photoshop). When it is saved in vector format itself, it is called a